

PROJECT ABSTRACT

Project Title: Azure Virtual Desktop Pooled Host Pool for an Engineering Lab
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Executive Abstract

Modern engineering education requires high-performance desktop computing for compute-intensive software suites (such as CAD, numerical modeling, finite element analysis, and integrated development environments). Providing dedicated physical workstations for every student incurs unsustainable capital costs, complex physical maintenance, and underutilization outside scheduled hours. This project implements an enterprise-grade, cloud-native **Azure Virtual Desktop (AVD) pooled host pool** designed to support a target cohort of approximately 50 students while operating strictly within student cloud credit limits.

Technical Investigation & Core Objectives

The project rigorously investigates two primary technical problem domains through live empirical testing rather than theoretical estimation:

- **Session-Host Sizing for Engineering Workloads:** Measuring CPU, memory, and disk utilization under multi-user concurrent loads to establish stable concurrency boundaries per virtual machine.
- **FSLogix Profile-Container Storage Scalability:** Evaluating whether Azure Files SMB storage (128 GiB, 1026 IOPS, 63 MiB/s) becomes an I/O bottleneck during simultaneous student logons (logon storms).
- **Enterprise Identity & Decoupling:** Implementing Microsoft Entra Kerberos authentication (AADKERB) and share-level RBAC for secure profile roaming across non-persistent multi-session VMs.
- **Cost Governance:** Enforcing strict budget safety through single-host testing, automated scheduled deallocation (18:00 UTC), and resource locking.

Architectural Summary

Layer	Deployed Azure Component	Configuration Details
Control Plane	AVD Host Pool, App Group, Workspace	Pooled, Breadth-First, Metadata anchored in centralindia
Network Security	vnet-avd-lab / snet-avd / nsg-avd-lab	10.0.0.0/16, Outbound TLS 443, SMB 445, KMS 1688
Compute	Session Host VM (Standard_B4ms)	Windows 11 Enterprise Multi-Session 23H2, 128 GB Standard SSD
Profile Storage	stavdprofiles260916 / fslogix share	Azure Files StandardV2_LRS, 128 GiB, 1026 IOPS, SMB 3.0
Identity / RBAC	AVD-Engineering-Lab-Students	Desktop Virtualization User & Storage File SMB Contributor

Expected Outcome

The project delivers a repeatable Infrastructure-as-Code deployment (Azure Bicep), verified student desktop connectivity, and empirical benchmark results (1, 3, and 5 concurrent users) establishing the exact session host count required for 50 enrolled students.